

Literature Review: Methods for Piecewise Signal Estimation

L_1 Trend Filtering, L_0 Trend Filtering, Optimal Partitioning/PELT, and SVP

1 Purpose

The objective of this review is to compare four approaches relevant to continuous piecewise-linear segmentation: L_1 trend filtering, L_0 trend filtering, Optimal Partitioning (OP)/PELT, and Smallest Valid Partitioning (SVP).

The main points of comparison are:

objective optimization structure computational complexity statistical control.

The final research objective is to extend SVP from piecewise-constant segmentation to continuous piecewise-linear signals. Therefore, particular attention is paid to methods that estimate changes in slope and to dynamic-programming formulations.

2 L_1 Trend Filtering

Given observations

$$y = (y_1, \dots, y_n)^\top,$$

k th-order trend filtering solves

$$\hat{\beta} = \arg \min_{\beta \in \mathbb{R}^n} \left\{ \frac{1}{2} \|y - \beta\|_2^2 + \lambda \|D^{(k+1)}\beta\|_1 \right\}.$$

For continuous piecewise-linear estimation, $k = 1$, so the penalty is applied to second differences:

$$(D^{(2)}\beta)_i = \beta_i - 2\beta_{i+1} + \beta_{i+2}.$$

Hence zeros in $D^{(2)}\beta$ correspond to regions of constant discrete slope, whereas nonzero entries identify changes in slope.

Convexity

The squared-error term

$$f(\beta) = \frac{1}{2} \|y - \beta\|_2^2$$

has Hessian

$$\nabla^2 f(\beta) = I_n \succ 0,$$

and is therefore strictly convex.

Since the L_1 norm is convex and $D^{(k+1)}$ is linear,

$$\beta \mapsto \|D^{(k+1)}\beta\|_1$$

is convex. Consequently, the complete objective is strictly convex and has a unique global minimizer.

The objective is nevertheless nonsmooth because

$$|z|$$

is not differentiable at $z = 0$. Thus ordinary gradient descent is not directly applicable at points where

$$(D^{(k+1)}\beta)_i = 0.$$

The first-order optimality condition is instead

$$0 \in \hat{\beta} - y + \lambda D^\top z,$$

where

$$z_i \in \begin{cases} \{1\}, & (D\hat{\beta})_i > 0, \\ [-1, 1], & (D\hat{\beta})_i = 0, \\ \{-1\}, & (D\hat{\beta})_i < 0. \end{cases}$$

Computation

One possible approach is a primal-dual interior-point method. Introducing $t_i \geq |(D\beta)_i|$ gives the equivalent constrained problem

$$\begin{aligned} \min_{\beta, t} \quad & \frac{1}{2} \|y - \beta\|_2^2 + \lambda \mathbf{1}^\top t \\ \text{s.t.} \quad & D\beta - t \leq 0, \quad -D\beta - t \leq 0. \end{aligned}$$

A primal-dual method applies Newton iterations to perturbed KKT conditions. After elimination of auxiliary variables, the main linear system has the form

$$(I + D^\top W D) \Delta\beta = r,$$

where W is diagonal.

For fixed trend-filtering order, D is sparse and banded; therefore $I + D^\top W D$ has fixed bandwidth. A Newton system can consequently be solved in

$$O(n)$$

time per iteration.

The classical worst-case interior-point bound is approximately

$$O\left(n^{3/2} \log \frac{1}{\varepsilon}\right)$$

for numerical accuracy ε , although practical behaviour may be substantially faster.

Statistical property and critique

Trend filtering has established nonparametric estimation guarantees. For functions satisfying

$$\text{TV}(f^{(k)}) \leq C,$$

an appropriately tuned estimator achieves the minimax rate

$$\frac{1}{n} \sum_{i=1}^n \left(\hat{f}(x_i) - f_0(x_i) \right)^2 = O_{\mathbb{P}} \left(n^{-\frac{2k+2}{2k+3}} \right).$$

For $k = 1$, this rate is

$$O_{\mathbb{P}}(n^{-4/5}).$$

The main advantages are convexity, automatic knot selection and strong estimation theory. The main limitation for the present project is that λ controls slope changes globally rather than imposing an explicit local validity condition. Furthermore, the L_1 penalty penalizes the magnitude of slope changes and can therefore shrink them.

3 L_0 Trend Filtering

The L_0 analogue is

$$\hat{\beta} = \arg \min_{\beta \in \mathbb{R}^n} \left\{ \frac{1}{2} \|y - \beta\|_2^2 + \lambda \|D^{(k+1)} \beta\|_0 \right\},$$

where

$$\|z\|_0 = \#\{i : z_i \neq 0\}.$$

For $k = 1$,

$$\|D^{(2)} \beta\|_0$$

counts nonzero second differences and therefore directly penalizes the number of slope changes.

Nonconvexity

The L_0 penalty is not convex. For example, in one dimension, choose

$$x = 0, \quad y = 1, \quad \theta = \frac{1}{2}.$$

Then

$$\left\| \frac{x+y}{2} \right\|_0 = 1 > \frac{1}{2} = \frac{1}{2}\|x\|_0 + \frac{1}{2}\|y\|_0.$$

Hence the convexity inequality is violated.

This leads to a combinatorial support-selection problem. If there are m possible nonzero locations and exactly q are selected, the number of possible supports is

$$\binom{m}{q},$$

while unrestricted enumeration contains

$$\sum_{q=0}^m \binom{m}{q} = 2^m$$

supports.

Unlike L_1 ,

$$\|cz\|_0 = \|z\|_0, \quad c \neq 0,$$

so L_0 penalizes whether a change exists rather than its magnitude.

AMIAS

Wen, Wang and Zhang propose the *Alternating Minimization Induced Active Set* (AMIAS) method.

For a prescribed cardinality q , the problem can be written

$$\boxed{\min_{\alpha} \frac{1}{2}\|y - \alpha\|_2^2 \quad \text{s.t.} \quad \|D\alpha\|_0 = q.}$$

The active set

$$A = \{i : (D\alpha)_i \neq 0\}$$

represents the estimated locations of derivative changes. AMIAS alternates between estimating the signal for a current active set and updating this set according to necessary optimality conditions.

For fixed trend-filtering order, its principal linear-algebra operations exploit the sparse structure of D , and the cost of an iteration can scale linearly in n . This does *not* imply that the complete nonconvex L_0 optimization problem is globally solved in $O(n)$ time.

Different cardinalities can be examined and a model-selection criterion, such as

$$\text{BIC} = n \log \left(\frac{\text{RSS}}{n} \right) + p \log n,$$

can be used to select model complexity.

Critique

L_0 has an attractive interpretation because it directly controls the number of changes and does not shrink their magnitudes. However, its nonconvex and combinatorial structure makes optimization more difficult than for L_1 . Specialized algorithms are therefore required, and a computed solution does not automatically inherit the global-optimum guarantees of a convex problem.

4 Optimal Partitioning and PELT

Optimal Partitioning addresses additive segmentation problems. Let

$$C_{s,t}$$

denote the cost of fitting observations $(s, t]$ as one segment and let $\beta > 0$ be a penalty per changepoint. Define

$$F(t) = \text{minimum penalized cost for observations } 1, \dots, t.$$

The dynamic-programming recurrence is

$$F(t) = \min_{0 \leq s < t} \{F(s) + C_{s,t} + \beta\},$$

with an appropriate initialization such as

$$F(0) = -\beta.$$

The recurrence is exact because, conditional on the last changepoint s , the optimal partition before s is independent of the final segment.

$O(1)$ segment costs

For piecewise-constant Gaussian segmentation,

$$C_{s,t} = \min_{\mu} \sum_{i=s+1}^t (y_i - \mu)^2.$$

Define cumulative sums

$$S_t = \sum_{i=1}^t y_i, \quad Q_t = \sum_{i=1}^t y_i^2.$$

Then

$$C_{s,t} = (Q_t - Q_s) - \frac{(S_t - S_s)^2}{t - s}.$$

After $O(n)$ preprocessing of S_t and Q_t , evaluating one segment cost requires only a fixed number of operations:

$$C_{s,t} = O(1).$$

Optimal Partitioning considers

$$\sum_{t=1}^n t = O(n^2)$$

candidate previous changepoints and therefore has complexity

$$O(n^2)$$

when segment costs are $O(1)$.

PELT improves this recurrence by pruning candidate changepoints that can be shown not to participate in an optimal future segmentation. The pruning is exact under its cost assumptions. Its worst-case complexity remains $O(n^2)$, while under suitable stochastic conditions its expected computational cost can be linear in n .

Critique

OP and PELT provide an exact global optimization interpretation and are fundamental references for segmentation by dynamic programming. However, model complexity is controlled through a global penalty β . Standard OP/PELT does not require every returned segment to pass an explicit local statistical validity test.

5 Smallest Valid Partitioning

SVP changes the optimization principle. Instead of selecting a global penalty on the number of changes, each returned segment must satisfy a local validity condition.

For a partition

$$0 = \tau_0 < \tau_1 < \dots < \tau_K = n,$$

SVP minimizes lexicographically

$$\boxed{\text{lex min} \left(K, \sum_{j=1}^K C_{\tau_{j-1}, \tau_j} \right)}$$

subject to

$$T_{\tau_{j-1}, \tau_j} \leq \gamma, \quad j = 1, \dots, K.$$

Thus the primary objective is to minimize the number of valid segments; the fitting cost resolves ties.

A possible cost-based validity statistic is

$$\boxed{T_{s,t} = \max_{s < u < t} \{C_{s,t} - C_{s,u} - C_{u,t}\}.$$

It measures the largest reduction in cost obtainable by introducing an internal split. A segment is considered valid if

$$T_{s,t} \leq \gamma.$$

Dynamic programming

Let R_t denote the optimal lexicographic value for observations up to t . The recurrence has the form

$$\boxed{R_t = \text{lex min}_{s < t} \{R_s + (1, C_{s,t}) : T_{s,t} \leq \gamma\}.$$

Therefore SVP remains an exact dynamic-programming method for its own optimization criterion.

Cost of validity checking

For the statistic above, evaluating one candidate segment $(s, t]$ requires examining each possible internal point

$$u = s + 1, \dots, t - 1.$$

For a segment of length $L = t - s$, this is

$$O(L)$$

work when individual costs are available in $O(1)$ time.

Since there are $O(n^2)$ possible segments, a direct implementation that recomputes validity independently can require up to

$$\boxed{O(n^3)}$$

operations.

Pruning and reuse of calculations can substantially reduce this cost, depending on the validity statistic and data.

Critique

SVP has an important property absent from the preceding methods:

$$\boxed{\text{every returned segment satisfies an explicit local validity test.}}$$

It also separates fitting cost from validity, which permits different statistical tests to be used without changing the fundamental optimization principle.

Its limitations are the need to select and calibrate γ , the potential cost of evaluating validity, and the dependence of computational performance on the chosen test and pruning properties.

6 Relevance to Continuous Piecewise-Linear SVP

The target model is

$$y_i = f(x_i) + \varepsilon_i, \quad x_1 < \dots < x_n,$$

where f is continuous and piecewise linear.

For a candidate segment $(s, t]$, let

$$u = f(x_s), \quad v = f(x_t).$$

Continuity gives the segment line

$$\ell_{s,t}^{u,v}(x) = u + \frac{v - u}{x_t - x_s}(x - x_s),$$

with squared-error cost

$$C_{s,t}(u, v) = \sum_{i=s+1}^t [y_i - \ell_{s,t}^{u,v}(x_i)]^2.$$

A natural extension of SVP is then

$$\text{lex min} \left(K, \sum_{j=1}^K C_{\tau_{j-1}, \tau_j}(z_{j-1}, z_j) \right)$$

subject to local validity conditions

$$T_{\tau_{j-1}, \tau_j}(z_{j-1}, z_j) \leq \gamma.$$

The central difficulty relative to ordinary OP or SVP is continuity: adjacent segments share the endpoint value z_j . Therefore the dynamic program must preserve a continuous state.

A natural state is

$$Q_t(z) = \text{optimal value for observations } 1:t \text{ subject to } f(x_t) = z,$$

leading formally to

$$Q_t(z) = \text{lex min}_{0 \leq s < t} \inf_u \{Q_s(u) + (1, C_{s,t}(u, z)) : T_{s,t}(u, z) \leq \gamma\}.$$

The main research questions are therefore whether this functional recurrence can be represented exactly, how the local validity region depends on (u, z) , and whether candidates can be pruned sufficiently to obtain an efficient algorithm.

7 Summary

The four methods address related but different optimization principles:

Method	Complexity control	Main property
L_1 TF	$\lambda \ D\beta\ _1$	convex, shrinkage
L_0 TF	$\lambda \ D\beta\ _0$	direct sparsity, nonconvex
OP/PELT	βK	exact segmentation DP
SVP	$T_{s,t} \leq \gamma$	explicit local validity

For continuous piecewise-linear SVP, the most relevant ideas are the slope-change representation of trend filtering, the exact dynamic-programming principle of OP/PELT, and SVP's separation between fitting cost and local validity. The principal new difficulty is the continuous endpoint state required to maintain continuity between adjacent linear segments.